SHREYAS YUVARAJ PATIL

AI/ML Engineer • Generative AI Engineer • Data Scientist

+91 7620280740 | shreyas.syp@outlook.com | linkedin.com/in/shreyaspatil | github.com/shreyaspatil | Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer and Generative AI Developer with 1+ years of professional experience designing and deploying production-grade ML, deep learning, and LLM-powered systems across aviation, railway, healthcare, and education domains. Expert in RAG pipelines, agentic AI workflows (LangChain, LangGraph, MCP), LLM fine-tuning (LoRA, QLoRA, PEFT), real-time voice AI, anomaly detection, and NLP. Delivered measurable impact: 20x ETL throughput improvement, 90% tokenization cost reduction, 2–3 week early fault detection, and sub-200ms real-time voice AI latency.

TECHNICAL SKILLS

Programming: Python (OOP, Asyncio), SQL (PostgreSQL, MySQL, Oracle, SQL Server), JavaScript, React
Generative AI & LLMs: LangChain, LangGraph, RAG (Hybrid Search, Re-Ranking), Agentic AI, Multi-Agent Systems, Prompt Engineering, Function Calling, Fine-Tuning (LoRA, QLoRA, PEFT), OpenAI API, Gemini API, Hugging Face Transformers, Embedding Optimization, MCP, Whisper STT
ML & Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, NLP, Computer Vision, Anomaly Detection, Time-Series Analysis, Signal Processing, Autoencoders, 1D-CNN, LSTM, ResNet, Isolation Forest, K-Means, DBSCAN, Power BI
Deployment & Infrastructure: FastAPI, Flask, Docker, Kubernetes, Redis, WebSockets, REST APIs, Microservices, CI/CD Pipelines, MLflow, Model Monitoring, Pydantic, Git, asyncio
Cloud & Vector Databases: AWS (S3, EC2), Microsoft Azure, Oracle Cloud (OCI), FAISS, Milvus, Pinecone, ChromaDB, Ollama

PROFESSIONAL EXPERIENCE

AI/ML Developer | Cloud Analogy CRM Specialist Limited — *Oct 2025 – Present*

IndigoVoiceAI – Real-Time Conversational Voice AI for IndiGo Airlines

Python, FastAPI, Ultravox, Azure GPT-4o Realtime, Redis, WebSockets, Silero VAD, asyncio, Function Calling, CI/CD, Model Monitoring

- Architected a production real-time voice AI system achieving sub-200ms end-to-end latency, integrating Ultravox STT, Azure GPT-4o Realtime LLM, and TTS across 3 concurrent engine configurations with automatic multi-engine failover (Ultravox → Azure GPT-4o → Azure Mini).
- Engineered a Redis-backed distributed session management layer with Silero VAD voice activity detection, ensuring zero-drop conversation continuity and zero-latency handoff across concurrent airline support calls.
- Built a per-turn model monitoring subsystem capturing 6 KPIs per exchange (latency, token cost, TTS duration, tool-call frequency, engine-switch rate, error rate) for real-time SLA monitoring and automated alerting via CI/CD deployment pipeline.
- Implemented function calling and tool calling integrations allowing the voice agent to query live flight status, booking APIs, and FAQ databases in real time during active customer calls.
- Automated end-to-end transcript processing pipeline — converting raw LLM history to structured logs with timing metadata, streaming JSON to an analytics server — reducing post-call manual logging by ~90%.

Energy7 (E7) – AI Pipeline for Railway Predictive Maintenance

Python, FastAPI, LangChain, LangGraph, PyTorch, Scikit-learn, Autoencoders, 1D-CNN, LSTM, Isolation Forest, Milvus, MLflow, AWS S3/EC2, Signal Processing, ETL, IoT

- Architected and shipped 5 production predictive-maintenance systems for railway infrastructure: track leakage detection, signal alert auditing, anomaly detection, point machine fault labeling, and a RAG-based documentation chatbot.
- Engineered an agentic LangChain orchestration pipeline processing high-frequency telemetry from 200+ Railway IXL Signaling units, achieving 99% signal compression and 90% reduction in LLM tokenization costs.
- Trained Autoencoder and 1D-CNN models for unsupervised fault detection on IoT time-series data; applied LSTM networks for TWS sequential signal analysis and K-Means/DBSCAN clustering to classify assets into 4+ health-state categories, enabling fault prediction 2–3 weeks ahead of failure.
- Tracked all model experiments, hyperparameter runs, and artifact versioning using MLflow; stored model artifacts and training datasets on AWS S3 with EC2 instances provisioned for compute-intensive training workloads.
- Applied Scikit-learn preprocessing pipelines for feature scaling, imputation, and cross-validation across multi-sensor telemetry streams; evaluated models using F1, precision-recall, and AUC-ROC metrics.
- Architected a hybrid RAG chatbot for railway engineering documentation using LangChain, LangGraph multi-agent orchestration, and Milvus vector search with semantic search and re-ranking.
- Delivered a 20x throughput improvement in the real-time ETL anomaly detection pipeline, directly reducing unplanned maintenance incidents across active railway assets.

Junior ML Engineer | Microsage Pvt Ltd — *July 2025 – Sep 2025*

GateTUTOR AI Learning Platform – Multi-Module Educational Intelligence System

Python, FastAPI, Flask, OpenAI API, RAG (FAISS + TF-IDF), Scikit-learn, SQL Server, PyTorch, Hugging Face Transformers, Docker, Kubernetes, Microservices, CI/CD, Pydantic

- Designed and deployed a production multi-module AI learning platform integrating MCQ generation, conversational chatbot, diagram rendering, plagiarism detection, and performance analytics within a unified microservices architecture via REST APIs.
- Developed a RAG-based MCQ generation pipeline using OpenAI APIs and FAISS vector retrieval over curated academic content — producing context-aware questions with configurable difficulty control and topic mapping.

- Implemented a hybrid retrieval chatbot (FAISS vector search + TF-IDF re-ranking) with Scikit-learn preprocessing pipelines, supporting multi-role access and improving retrieval precision across academic workflows.
- Built a multi-strategy code plagiarism detection engine (AST structural comparison + token n-gram Jaccard + exact-match fingerprinting), achieving 95% precision on a 3,000-submission validation set with <2% false positive rate.
- Containerized all modules with Docker and orchestrated deployments using Kubernetes; established CI/CD pipelines for automated model serving and zero-downtime rollouts across internal exam and learning systems.

INTERNSHIPS

AI/ML Intern | **EdTech Society, IIT Bombay** *Apr 2024 – Sep 2024*

- Trained a ResNet-50 facial emotion recognition model on 50,000 DAiSEE video samples, achieving 88% multi-class accuracy across 4 engagement states using transfer learning, data augmentation, and dropout regularization.
- Labeled and preprocessed 10,000+ video frames via Python/OpenCV pipelines; deployed a real-time inference pipeline processing 12 FPS live video for classroom engagement monitoring.

Machine Learning Intern | **Prarambh Development Cell** *Sep 2023 – Jan 2024*

- Developed a CNN Alzheimer's stage classifier using TensorFlow/Keras, trained on 5,000 MRI images with data augmentation and batch normalization, achieving 87% validation accuracy; deployed as a Flask REST API for real-time medical inference.

Data Analyst Intern | **Simdaa Technologies** *Jun 2023 – Jul 2023*

- Built Power BI dashboards processing 100,000+ records; optimized SQL queries (full scans → indexed joins) and rewrote DAX measures, reducing dashboard load time by 30% and enabling real-time KPI monitoring.

PERSONAL PROJECTS

OCR Fine-Tuning Research – CAPTCHA & Hindi Text Recognition | *Python, Tesseract5, TrOCR, LoRA/QLoRA (PEFT), Hugging Face, PyTorch, Ollama, ChromaDB, Quantization*
GitHub: tesseract5-captcha-finetuning | fast-hindi-ocr-extractor

- Fine-tuned Tesseract5 engine on synthetic CAPTCHA datasets and TrOCR (Hugging Face Transformers) for Hindi document extraction using QLoRA (PEFT) with 4-bit quantization, reducing GPU memory footprint by ~60% while preserving recognition accuracy.
- Built a local inference pipeline using Ollama for model serving and ChromaDB for embedding-based document retrieval; applied Pinecone for scalable vector indexing in multi-document search experiments.

InnerTone – AI Mental Wellness Consultation Platform | *Python, FastAPI, React (Three Fiber), WebRTC, Gemini API, PostgreSQL, RAG, Embedding Optimization*

- Built a full-stack AI mental wellness platform with real-time voice/video/text consultations via WebRTC and WebSockets, featuring a React-powered 3D avatar with real-time lip-sync and a RAG knowledge retrieval system grounded in CBT/psychology resources.
- Optimized embedding strategies for the CBT knowledge base — benchmarked chunking sizes, embedding models (text-embedding-3-small vs. large), and retrieval configurations to maximize semantic precision for mental health query resolution.
- Implemented AI safety guardrails for automated detection, classification, and escalation of high-risk conversations (suicidal ideation, emotional distress) — demonstrating responsible AI development and LLM safety practices.

SideKick – Intelligent Job Application Assistant | *JavaScript, React, Chrome Extensions API, Python, REST APIs, Web Scraping, AI Resume Parsing*

- Built a Chrome Extension for one-click job applications across 10+ platforms with an AI-powered job aggregation pipeline, resume parsing, candidate profiling, and secure REST API backend for multi-portal application management.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 – Generative AI Professional — Oracle
- Oracle Cloud Infrastructure 2025 – AI Foundations Associate — Oracle
- Oracle Cloud Infrastructure 2025 – Data Science Professional — Oracle
- Fundamentals of Deep Learning — NVIDIA Deep Learning Institute
- Python for Data Science, AI & Development — IBM / Coursera
- Data Analytics Essentials — Cisco Networking Academy

EDUCATION

B.Tech – Computer Science Engineering (AIML) *2021 – 2025*
DY Patil College of Engineering and Technology, Kolhapur | CGPA: 7.5 / 10